

SIMARPREET SINGH

STUDENT SOFTWARE DEVELOPER

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Education

Queen Mary's Grammar School, Walsall
Year 9 Student

Personal Profile

Software developer based in Walsall, UK, with a focus on systems programming, compiler development, web development, and self-hosted infrastructure. Creator of multiple independent projects spanning systems programming, language design, and full-stack development, alongside a freelance web design business serving small businesses across the West Midlands. Passionate about open-source software, self-hosted infrastructure, and continuous learning.

TECHNICAL SKILLS

Languages

- Python
- x86-64 Assembly (NASM)
- PHP
- Java
- SQL

Systems & Web

- Linux administration
- Compiler & language design
- Networking (P2P, DHT)
- WebAuthn / passkeys
- Responsive web design
- HTML / CSS

Tools & Infrastructure

- Git & GitHub
- Docker
- Nginx
- Cloudflare
- MariaDB / MySQL
- KVM / libvirt

SELECTED PROJECTS

Repositories linked from the portfolio above. Four independent projects spanning systems programming, languages, and full-stack development.

TeapotOS

Open Source · Ongoing (Aug 2026–present)

[Assembly \(NASM\)](#) · [x86](#) · [Bootloaders](#) · [Kernel Dev](#)

- Hand-wrote a bootloader and 32-bit kernel from scratch in x86 assembly - from nothing to a booting kernel in 2 days (~11KB of assembly)
- Implemented the real-mode to protected-mode transition: GDT setup, segment configuration, and VGA text buffer output
- Boots to a working interactive shell in QEMU

TeapotLang

Open Source · Ongoing (Jan 2024–present)

[Python](#) · [Compilers](#) · [AST](#) · [CI/CD](#)

- Designed and built a memory-efficiency-oriented programming language and compiler from the lexer and parser upwards
- 190+ commits; grew the integration/unit test suite to 9,000+ tests at 100% code coverage across the compiler pipeline and CLI
- Automated CI with ruff, mypy, CodeCov, and pytest
- Built a companion project website with an in-browser compiler demo (Pyodide/WASM) and a releases page

Tenberal News

Proprietary · Live

[PHP](#) · [MariaDB](#) · [Linux](#) · [Nginx](#) · [WebAuthn](#)

- Designed and built the backend of a CRUD news platform - authentication, article management, and reporter management (frontend by a separate developer)
- Implemented WebAuthn passkey authentication, including cross-browser and iOS Safari compatibility fixes
- Built a football World Cup statistics dashboard integrating the football-data.org API
- Self-hosted and administered the deployment on Linux with PHP, MariaDB, and Nginx

Portfolio Site

Open Source · Live

[HTML](#) · [CSS](#) · [JS](#) · [GitHub API](#)

- Implemented a fully static portfolio website using HTML, CSS, and JavaScript.
- Integrated the GitHub API to display project versions and contribution activity.
- Wrote detailed case studies for projects and embedded interactive project demos directly into the site.
- Created live previews of freelance websites using iframe embeds.

FREELANCE / CLIENT WORK

Run a freelance web design business building websites for small businesses across the West Midlands - salons, cafés, and retailers - that have no existing web presence.

Alisha's Beauty Salon <i>Walsall</i> Responsive informational site giving the salon its first web presence, built and launched in 2 weeks. <i>HTML · CSS · Responsive Design</i> alishasbeautysalon.netlify.app	Alisha Supermarket <i>Wolverhampton</i> Landing page for a Polish goods store, delivering a clear first online presence within 2 weeks. <i>HTML · CSS · Responsive Design</i> alishasupermarket.netlify.app
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- Built a lead-generation pipeline using Google Places data, processed into CRM-ready spreadsheets for prospecting
- Built a PHP tool (generator.php) to automate generation of client demo websites, replacing a manual bash-script workflow
- Worked with Google Ads and created a landing page for potential clients to read about my services

Technical Experience • Self-hosted Linux servers, incl. Docker container deployment and management • Nginx web server configuration and Cloudflare DNS/domain management • Git and GitHub workflows across every project	Leadership & Public Speaking • Invited to deliver a speech to officials and representatives involved in UK-China relations • Delivered the speech bilingually, in both Mandarin and English
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LANGUAGES

Punjabi Native speaker	English Native speaker	Mandarin Intermediate proficiency <i>Mandarin Excellence Programme (MEP) · 1st place in national Mandarin competition</i>
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